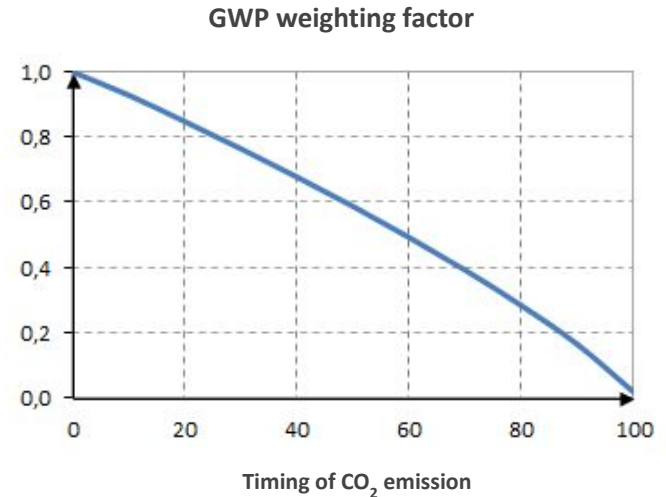




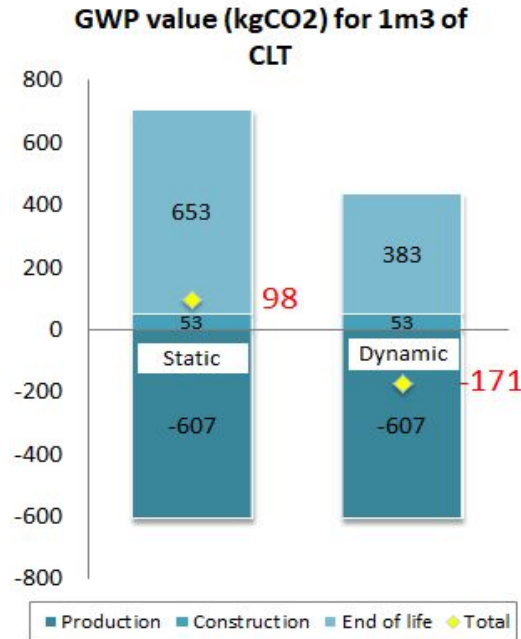
RE2020 / Simplified dynamic LCA

Introduction to RE2020 environmental regulation in France

- RE2020 regulation aims to replace the current thermal regulation (RT2012). Its application has been pushed twice from summer of 2020, to summer of 2021 and recently to January 2022
- The RE2020 regulation sets step by step CO₂ reduction targets (compared to the current situation) :
 - 15% de kgCO₂eq/m² in 2025, -25% in 2028, and from -30 to -40% in 2031.
- However, one of the biggest changes relates to the use of a simplified dynamic LCA approach for GWP calculation based on a 100 yr period since construction. This methodology essentially discounts emissions at later stages



Impact of RE2020 simplified dynamic LCA methodology



- Major influence from the credit for temporal carbon storage
- Advantage for wood products with significant emissions at the end of life

Note: the result of static LCA is extracted from French EPD of concrete and CLT. For this, it was considered that the main GHG is carbon dioxide

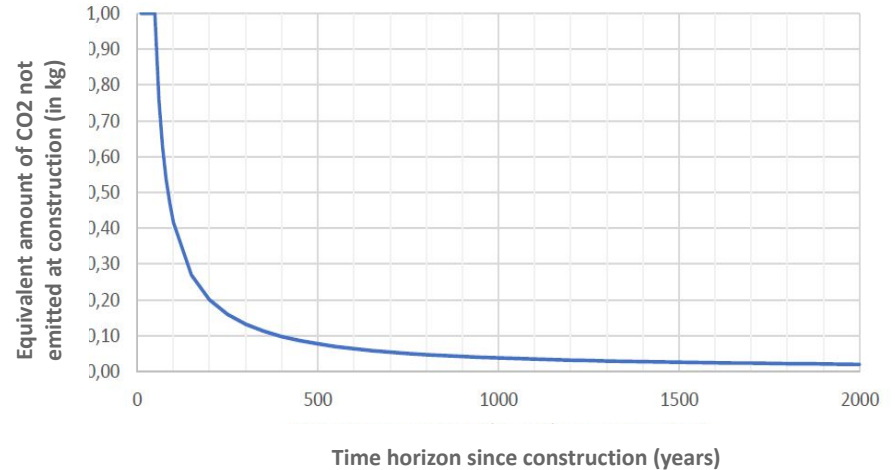
Stakeholder reaction - Traditional material suppliers

- Alongside the cement and concrete industry, several other industry associations are also against the dynamic LCA methodology ; minerals and metals, aluminium, insulation, bricks and tiles and plastics



Key arguments of traditional material suppliers (1/2)

- Dynamic LCA has methodological limitations. Its use shows that temporary carbon storage does not mitigate climate change
- Simplified dynamic LCA does not respect international consensus on excluding temporary carbon storage (EN 15804, PEF, ISO 14067)
- The results of the simplified dynamic LCA methodology depend strongly on the time horizon
 - 100 years after construction: storing 1 kg of CO₂ for 50 years is equivalent to not emitting 0.42 kg of CO₂ during construction;
 - This is reduced to 0.27 kg CO₂ at 150 years horizon (e.g. 100 years since emission as traditionally accounted)



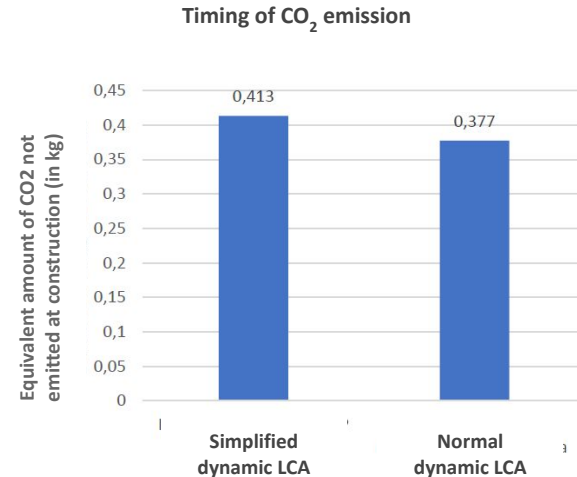
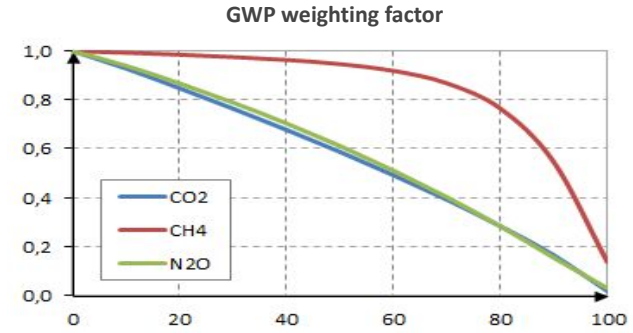
Key arguments of traditional material suppliers (2/2)

- The weighting factors used do not differentiate between the different greenhouse gases (CO_2 , CH_4) and thus it underestimates impacts from gases with strong GWP potential but short lifetimes (e.g. CH_4)

This is due to the fact that the calculation is based on existing aggregated GWP results from EPDs.

- All timber EPDs take into account a carbon credit linked to the fact that another tree will grow back on the plot from which the tree was taken, and that it will capture carbon.

Although this CO_2 capture will take 10-30 years, no GWP weighting factor is applied to discount the timing of this benefit (bottom right graph is for a 10 year growth period)



Stakeholder reaction - suppliers of wood-based products

- In June 2019 several timber associations sent a letter to DHUP (Department of Housing, Town Planning and Landscapes) to support the simplified dynamic LCA approach.
- In particular, this letter mentions:

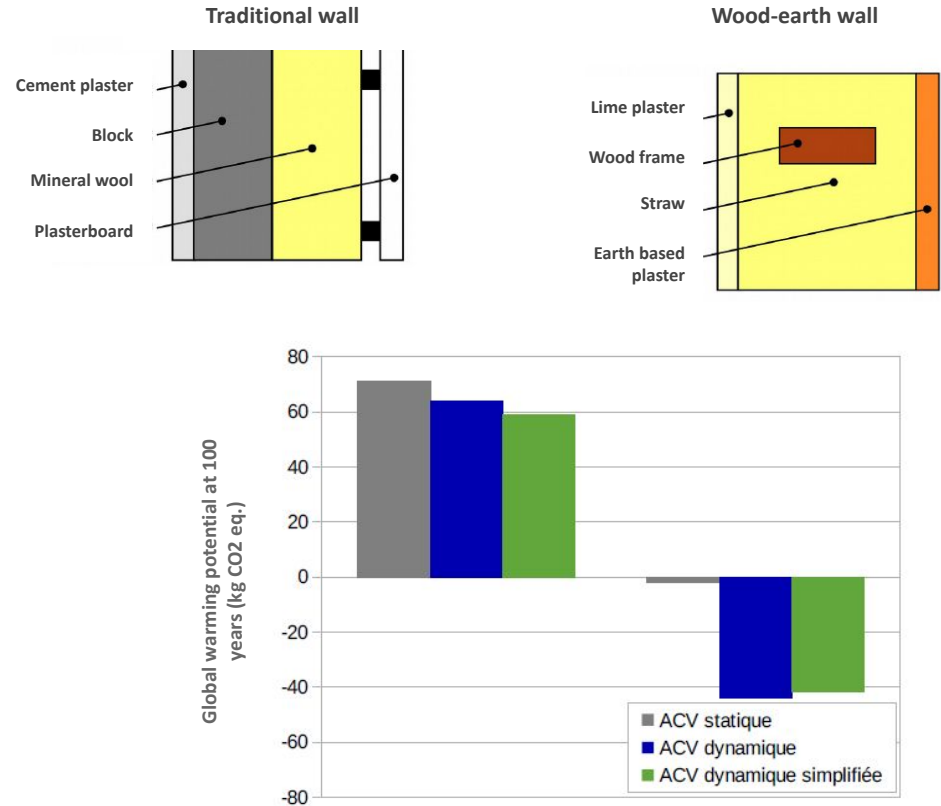
“Not taking this delay into account amounts to ignoring the benefit of storage in biobased materials: throughout the life of the building, the carbon stored temporarily does not contribute to climate change.

This approach is relevant because it is based on physical principles, it does not add any additional complexity to the current method and is operational today with the data present in the LCA databases.”



Key arguments of wood based material suppliers (1/2)

- Their argumentation is based on an example for a wall build with traditional materials or with wood-base products
- Based on this example, they claim that:
 - static LCA overestimates the real impact of construction methods with bio-based materials. The approach used previously is penalizing for bio-based sources and it does not represent reality.
 - simplified dynamic LCA gives results very close to dynamic LCA, and therefore to reality, for the two construction modes.
- It is important to note that (i) static LCA already gives negative CO₂ results for wood-based products (due to EoL scenario and landfill assumptions) and (ii) the analysis does not consider the time horizon issue



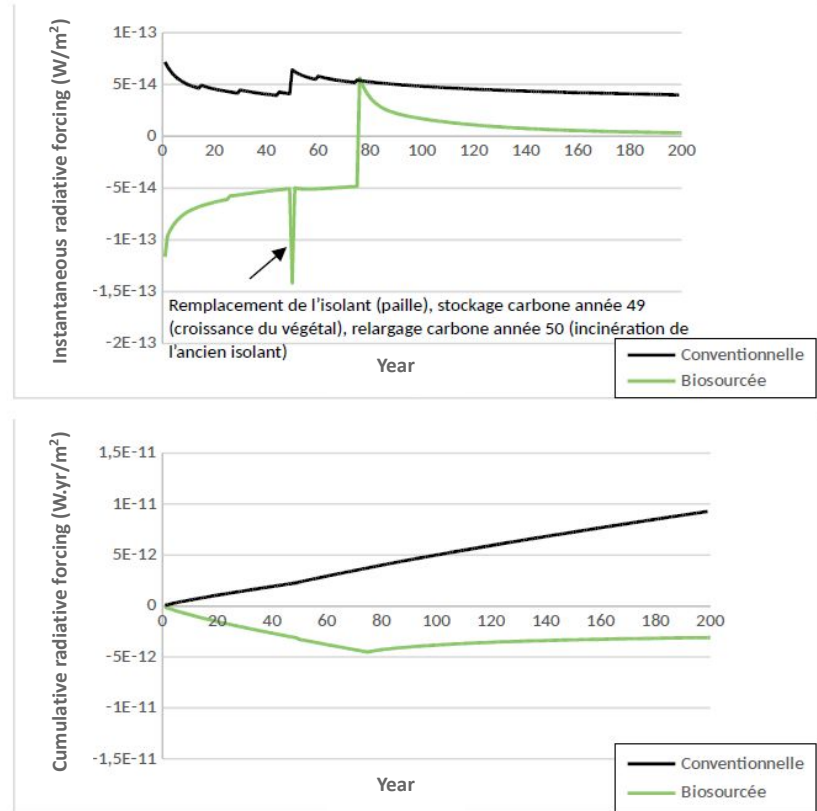
Key arguments of wood based material suppliers (2/2)

- The letter tries to refute the argument that
"Taking into account the issuance delay would only postpone the problem to future generations."

To do so, they show in the appendix results on instantaneous and cumulative radiative forcing and claim that:

"It is shown in the appendix to this letter that temporarily storing carbon represents a gain whatever the time horizon. The increase in use of biobased sources does not imply any risk of emissions being transferred to future generations."

- No real evidence is provided for the impact of scaling up production of wood-based materials.



Stakeholder reaction - contractors and other players

- The Higher Construction Council (CSCEE) also pled for the establishment of "a review clause on a closed list of requirements and methodological choices, and called for a return to the "standardized" LCA approach used previously
- The Effinergie association (includes players from different backgrounds have come together to promote energy efficiency in buildings) is also opposed to the use of the dynamic LCA method as it is inducing "major distortions between the sectors", and is imposed "outside any prior experimentation"
- The use of simplified dynamic LCA seems to be viewed positively by constructors who anyway concentrate more on their concerns with the RE2020 regulation (e.g. how different types of energy are accounted in the new regulation)



Overall

- To date, there is no evidence of the French government having carried out an impact assessment of the RE2020 regulation
- There is also no evidence that the government is changing position on the use of the simplified dynamic LCA methodology.
- However, it is claimed in the media that:
“The government has significantly moderated its speech compared to the announcements of November 2020: by 2030, the State no longer advances to say that the wooden frame will have become the standard in single-family homes. Thus, the thresholds for carbon emissions from materials have been slightly increased, according to the new 2025-2028-2031 calendar, enough to “promote the diversity of construction methods”.
- The French government has also announced on the 18th of February its plans to launch normalisation proceedings for the dynamic LCA methodology